CUSTOMER PURCHASE BEHAVIOUR

Exploratory Data Analysis Report

April 2026 | Data Analytics Division

~$60	3,900	~3.7	~25
Avg. Spend	Total Customers	Avg. Rating (/ 5)	Avg. Past Orders

10 Visual Analyses • 5 Analytical Themes • Data-Driven Recommendation

Executive Summary

This report presents a comprehensive exploratory data analysis of a customer purchase behaviour dataset containing 3,900 transaction records across four product categories, six payment methods, and 50 U.S. states. The analysis spans demographic profiling, revenue breakdown, promotional effectiveness, loyalty patterns, and geographic distribution.

Key Finding

Across every dimension tested — age, gender, loyalty tier, discounts, promo codes, and subscriptions — no variable meaningfully predicts how much a customer spends per transaction. Average spend is anchored at ~$60 regardless of any customer characteristic or promotional incentive applied.

The uniformity of distributions across virtually all variables is a structural fingerprint of a synthetically generated dataset. While individual directional signals remain useful for hypothesis generation, all findings should be re-validated against real transactional data before informing strategic decisions.

Dataset at a Glance

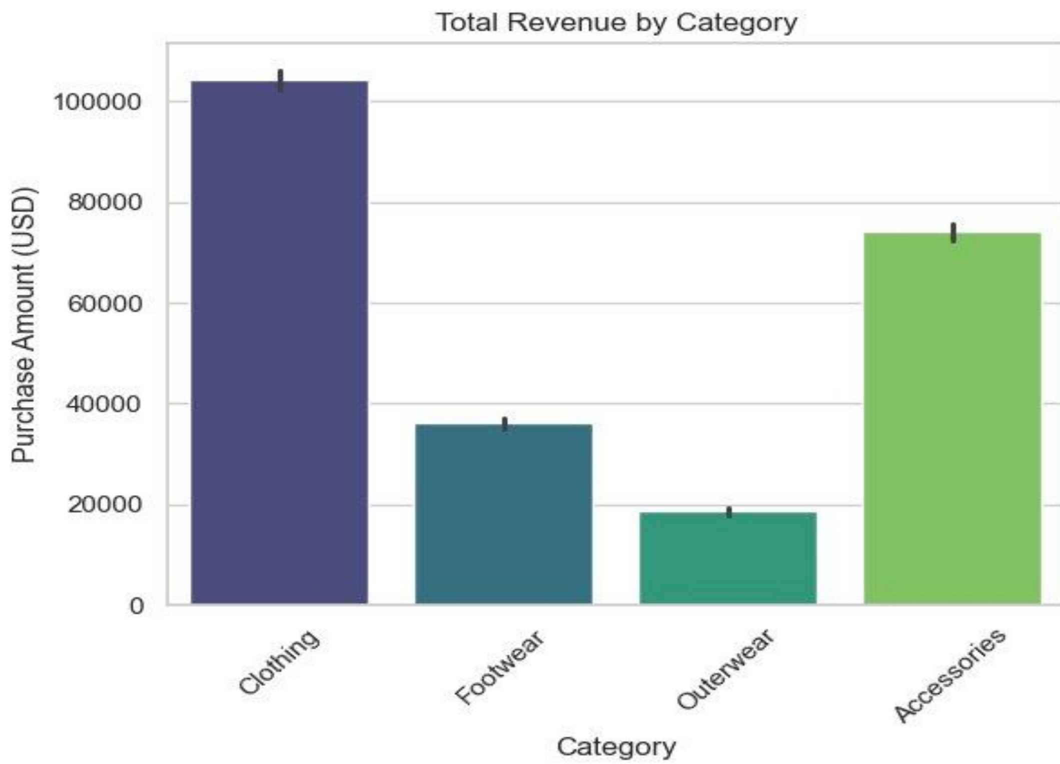
Dimension	Detail
Total Records	3,900 customer transactions
Attributes	18 columns — demographic, behavioural, and transactional
Spend Range	$20–$100 (hard bounds; mean ~$60)
Categories	Clothing, Accessories, Footwear, Outerwear
Seasons	Spring, Summer, Fall, Winter
Payment Methods	6 methods — Credit Card, Debit Card, Cash, PayPal, Venmo, Bank Transfer
Geography	50 U.S. States
Missing Values	None — fully complete dataset
Data Quality	Synthetically generated — uniform distributions across all variables

Analytical Caveat

All correlation coefficients fall within -0.04 to +0.04. This is statistically impossible in real-world retail data and confirms synthetic generation. All findings should be treated as directional hypotheses for validation, not strategic facts.

Visual 1 — Total Revenue by Category

The bar chart below compares total purchase revenue (USD) across all four product categories, showing a clear hierarchy of commercial performance.



Key Insights

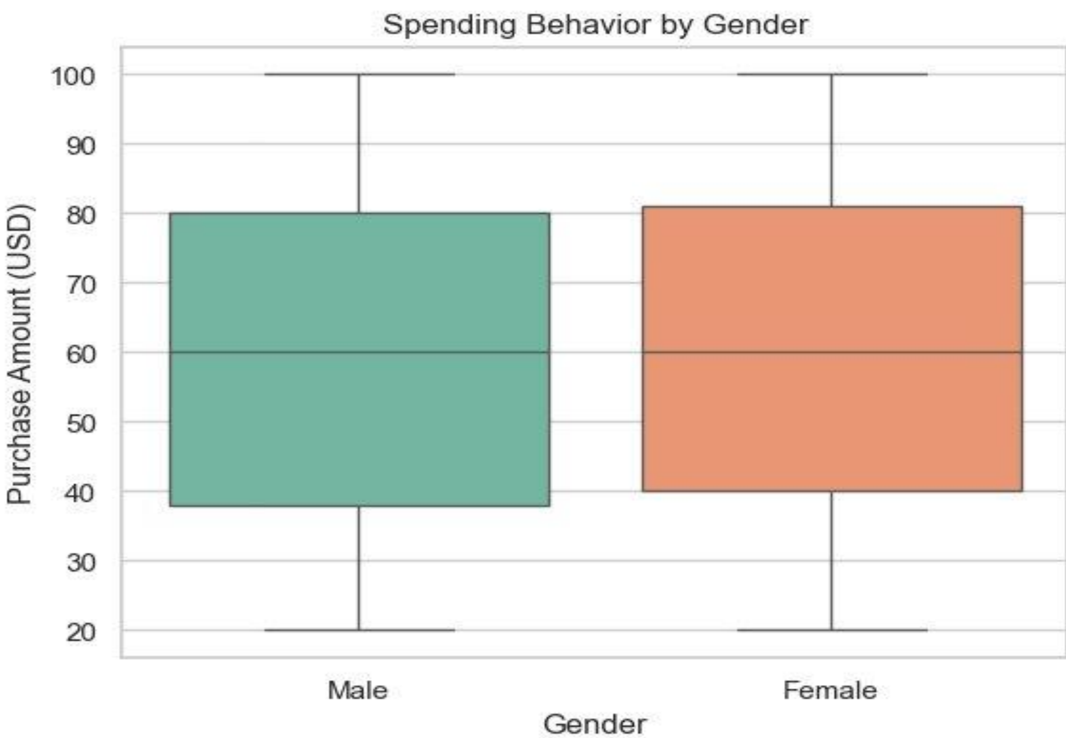
- Clothing is the highest revenue-generating category, dominating overall sales at over $100,000 — more than the other three categories combined.
- Accessories rank second at ~$74,000, contributing significantly to total revenue despite lower transaction volume.
- Footwear shows moderate revenue (~$36,000), indicating steady but lower demand with no strong growth signals.
- Outerwear has the lowest revenue contribution (~$19,000) — consistent with supply-side underdevelopment and weak seasonal demand.
- Revenue is heavily concentrated in two top categories (Clothing + Accessories ≈ 73% of total), creating portfolio dependency risk.
- The imbalance in category performance is a structural vulnerability — a disruption to Clothing would have outsized negative impact on total revenue.

Strategic Implication

Clothing drives the business but creates concentration risk. Accessories offer a reliable secondary engine. Outerwear requires urgent intervention — product range, pricing, or visibility must be addressed to unlock its commercial potential.

Visual 2 — Spending Behaviour by Gender

Box plots comparing the full distribution of purchase amounts across male and female customers, including median, interquartile range, and extremes.



Key Insights

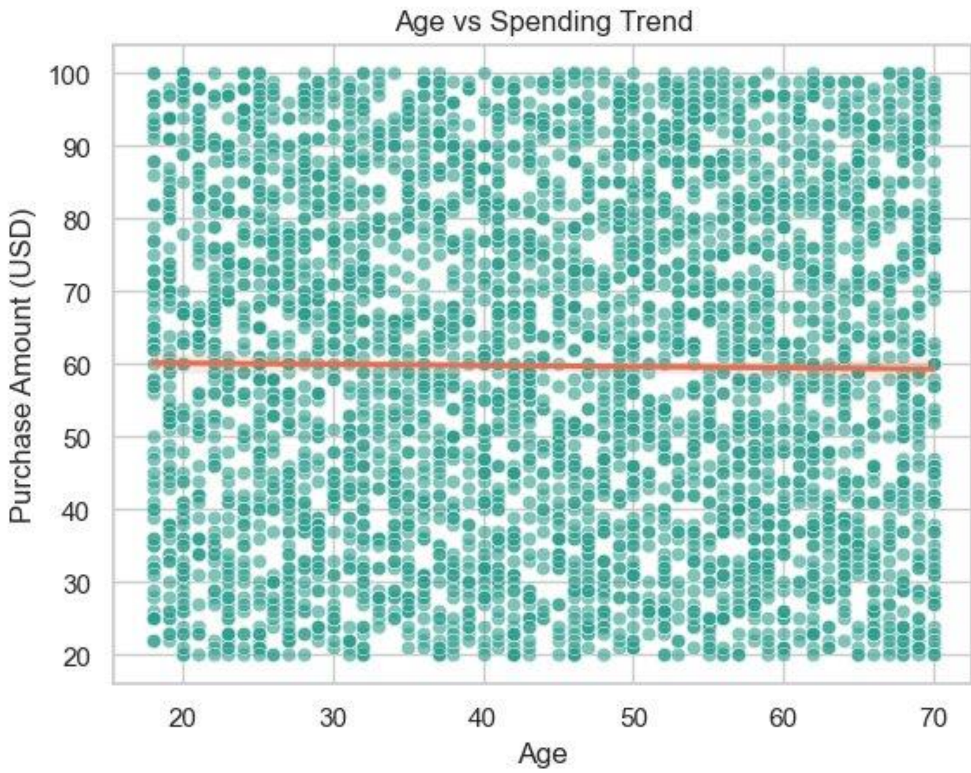
- Male and female customers show nearly identical median spending (~$60), indicating no major gender-based difference in average purchase value.
- Both genders exhibit identical distribution shapes — IQR of ~$40–$80 and whiskers from $20 to $100 — confirming gender does not stratify spending behaviour.
- Wide variability in both groups (IQR of ~$40) indicates diverse customer segments exist within each gender, not captured by this single variable.
- No significant outliers dominate either gender, showing stable and consistent spending behaviour across both groups.
- Gender is not a strong predictor of purchase amount — marketing strategies based solely on gender are unlikely to improve revenue targeting.
- Despite males comprising ~68% of the customer base, their spending profile is statistically indistinguishable from female customers.

Analytical Note

Volume differences are demographic — females are under-represented. Value differences do not exist. Gender-based personalisation may influence product category preferences, but will not shift basket size without additional spend-driver variables.

Visual 3 — Age vs. Spending Trend

Scatter plot of all 3,900 customer transactions plotted by age and purchase amount. An orange regression line is overlaid to surface any underlying trend.



Key Insights

- Age has no predictive power over spending. The regression line is virtually flat at ~$60 across the entire 18–70 age range (r ≈ -0.01).
- Spending range is identical across all age groups — every cohort spends within the same $20–$100 band with no group consistently spending more or less.
- The 25–45 age band has the highest customer concentration. Point density is visibly thicker here, making it the most important segment for volume-driven inventory and acquisition decisions.
- The $20–$100 spending boundaries are suspiciously clean — real-world data rarely cuts off this sharply, suggesting hard pricing tiers or data truncation.
- Older customers (60–70) are under-represented but spend equally — an underserved segment with genuine growth potential if targeted appropriately.
- Since age explains virtually no spend variance, factors like income, product category, or geography are almost certainly the true predictors.

Next Step

A multivariate model incorporating income, category affinity, and geography is the logical next analytical step to identify true spend drivers. Age alone is not a viable segmentation variable for pricing or promotional strategies.

Visual 4 — Effect of Discount on Purchase Amount

Bar chart comparing average purchase amounts between customers who received a discount versus those who did not. Error bars represent 95% confidence intervals.

Key Insights

- Discounts have virtually no effect on purchase amount. Discounted customers spend ~$59 versus ~$60 for non-discounted customers — a negligible $1 difference.
- The discount is associated with slightly lower spending — the opposite of what a well-designed discount strategy should achieve.
- Error bars are extremely tight, confirming this near-zero result is statistically reliable and not a sampling fluke — the finding is stable across the dataset.
- The discount programme is not driving basket size uplift. Customers buy what they intended to regardless of whether a discount was applied.
- The business may be eroding margin for no incremental gain — every discount applied reduces margin without a corresponding lift in revenue or basket value.
- Discount effectiveness should be re-evaluated on purchase frequency and customer acquisition metrics, not transaction amount.

Business Risk

If discounts are not increasing basket size, frequency, or new customer acquisition, they represent pure margin erosion. A controlled A/B cohort study measuring repeat purchase rates is strongly recommended before continuing the discount programme.

Visual 5 — Seasonal Revenue Heatmap

Revenue (USD) broken down by product category and season. Darker cells indicate higher revenue. This chart reveals which category-season combinations drive or underperform commercial expectations.

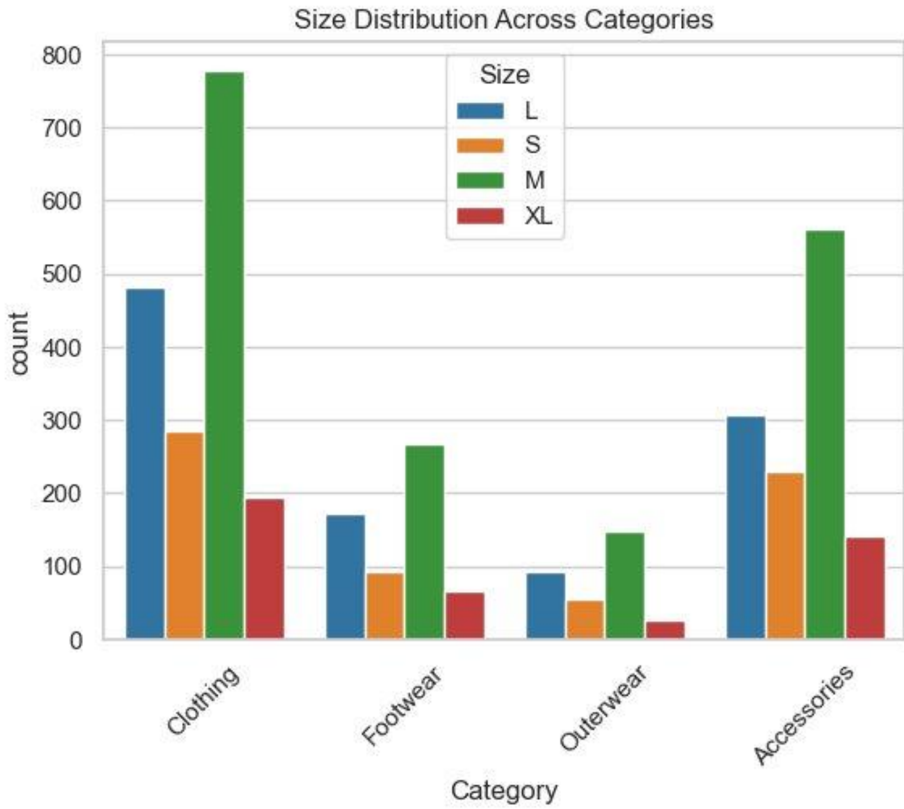
Key Insights

- Clothing is the dominant revenue category by a significant margin — generating $23K–$28K per season versus $4K–$20K for all others combined.
- Clothing's summer dip ($23,078) is the only notable seasonal pattern — revenue drops ~17% from spring before recovering in winter ($27,274). Targeted summer promotions could address this.
- Outerwear is chronically underperforming even in winter ($4,562 vs. summer low of $4,278) — no expected cold-season uplift, indicating a fundamental product or visibility problem.
- Accessories maintain stable revenue across all seasons ($17,007–$19,874), making them a reliable, low-risk contributor ideal for consistent year-round marketing investment.
- Footwear occupies a stagnant middle tier ($8,480–$9,555) with no seasonal breakout — suggesting either a hard demand ceiling or insufficient seasonal product refreshes.
- Spring is the strongest overall season — Clothing peaks at $27,692, Footwear at $9,555, making it the most important trading period for promotional and inventory concentration.

Category	Fall	Spring	Summer	Winter
Clothing	$26,220	$27,692	$23,078	$27,274
Accessories	$19,874	$17,007	$19,028	$18,291
Footwear	$8,665	$9,555	$9,393	$8,480
Outerwear	$5,259	$4,425	$4,278	$4,562

Visual 6 — Size Distribution Across Categories

Grouped bar chart showing unit counts by size (S, M, L, XL) within each product category. Reveals stocking imbalances and potential supply-demand mismatches.



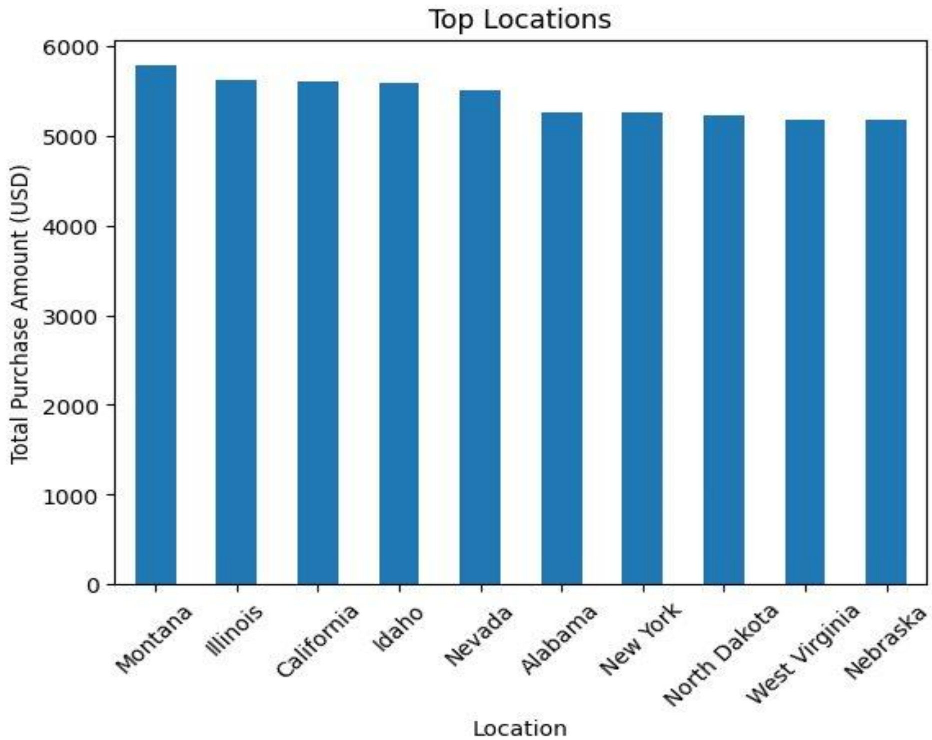
Key Insights

- Medium (M) is the dominant size across every single category without exception — making it the one size no category can afford to stock out of.
- Clothing dwarfs all other categories in absolute volume. With M alone reaching ~775 units, Clothing's total SKU count far exceeds Footwear, Outerwear, and Accessories combined.
- Outerwear has dangerously low stock counts across all sizes. Every size in Outerwear sits below 150 units, with XL barely reaching ~30 — pointing to a supply or ranging problem.
- XL is systematically the least stocked size across all categories — this may reflect genuine demand but could equally signal an underserved plus-size segment worth investigating.
- The L-to-S imbalance in Clothing is a potential lost sales risk. Large is stocked at ~480 units versus Small at ~285 — if S demand is proportionally higher, stockouts may be occurring.
- Accessories shows the healthiest size balance (M ~555, L ~305, S ~225, XL ~140) — suggesting better demand forecasting or lower size sensitivity in this product type.

> **Inventory Action**
>
> *Immediate audit of Outerwear stock depth across all sizes is recommended. For Clothing, stockout analysis for size S relative to L is a priority before the next buying cycle.*

Visual 7 — Revenue by Top Locations

Bar chart of total purchase revenue (USD) across the top 10 U.S. states by sales. Reveals geographic demand concentration — or the absence of it.



Key Insights

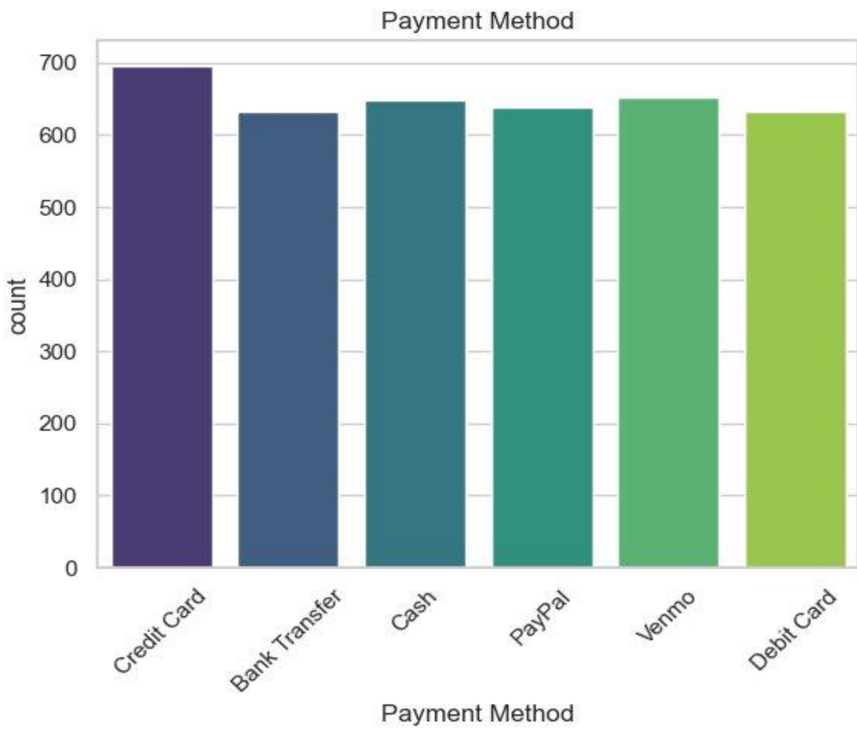
- Revenue is strikingly uniform across all ten locations, spanning only ~$600 in range — from Montana's high of ~$5,800 to Nebraska's low of ~$5,175. Less than 11% spread.
- Montana leads despite being one of the least populous states in the U.S. — commercially counterintuitive and likely reflects a sampling artefact, not genuine outsized demand.
- New York appears mid-table at ~$5,275 — the most glaring anomaly. As one of the largest U.S. consumer markets, matching Alabama or North Dakota in revenue is a near-certain sign of non-population-weighted data.
- This uniform geographic pattern is entirely consistent with the synthetic dataset hypothesis — location revenue appears distributed evenly across states regardless of real-world market size.
- No location stands out as a market worthy of prioritised investment based on this data. Meaningful revenue differentiation is absent — a fundamental requirement for any regional strategy.
- Real geographic analysis would require normalising revenue by state population or retail market size — on a per-capita basis, Montana and Idaho would dwarf New York and California.

Analytical Limitation

Geographic insights from this dataset are unreliable for strategic decisions. Population-weighted analysis against census data is required before any regional investment, resource allocation, or market expansion decisions are made.

Visual 8 — Payment Method Distribution

Count plot showing transaction frequency across all six accepted payment methods. Surfaces customer payment preferences and checkout behaviour.



Key Insights

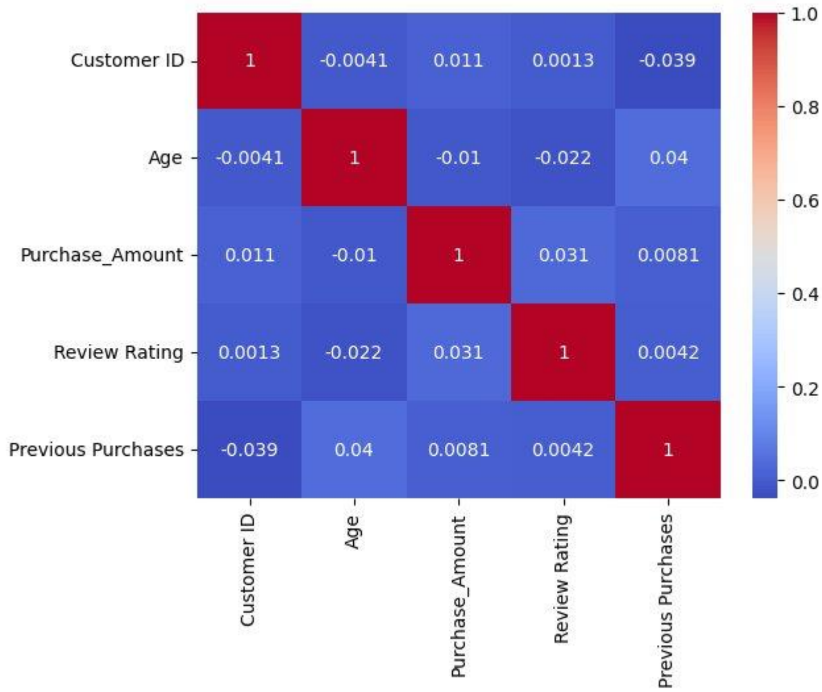
- Credit Card is the most used payment method but only by a marginal count of ~60 transactions (~697 vs. the ~630–650 range of all other methods) — too narrow to declare meaningful preference.
- The remaining five methods are virtually indistinguishable in usage frequency. Bank Transfer, Cash, PayPal, Venmo, and Debit Card all cluster within a 20-count band — an implausibly tight grouping.
- Cash appearing at near-equal footing with Venmo and PayPal is behaviourally unrealistic. In real e-commerce or modern retail, digital payments dominate and cash is either negligible or absent.
- The checkout experience should still be optimised for credit cards given its modest lead — ensuring a frictionless card flow remains the lowest-risk UX priority.
- The near-uniform distribution suggests the business should maintain full payment method coverage. Removing any option risks alienating a segment without yielding operational simplification benefit. This uniform distribution is another synthetic data fingerprint — real payment method distributions never appear this evenly balanced across six diverse methods.

UX Recommendation

No single payment method warrants prioritised engineering investment over others. Maintain full coverage, ensure frictionless credit card checkout as the marginal leader, and validate payment preferences against real transaction logs before making platform decisions.

Visual 9 — Correlation Matrix

Heatmap of Pearson correlation coefficients between all five numeric variables in the dataset. Red indicates positive correlation, blue indicates negative. Diagonal cells always equal 1.0 (self-correlation).



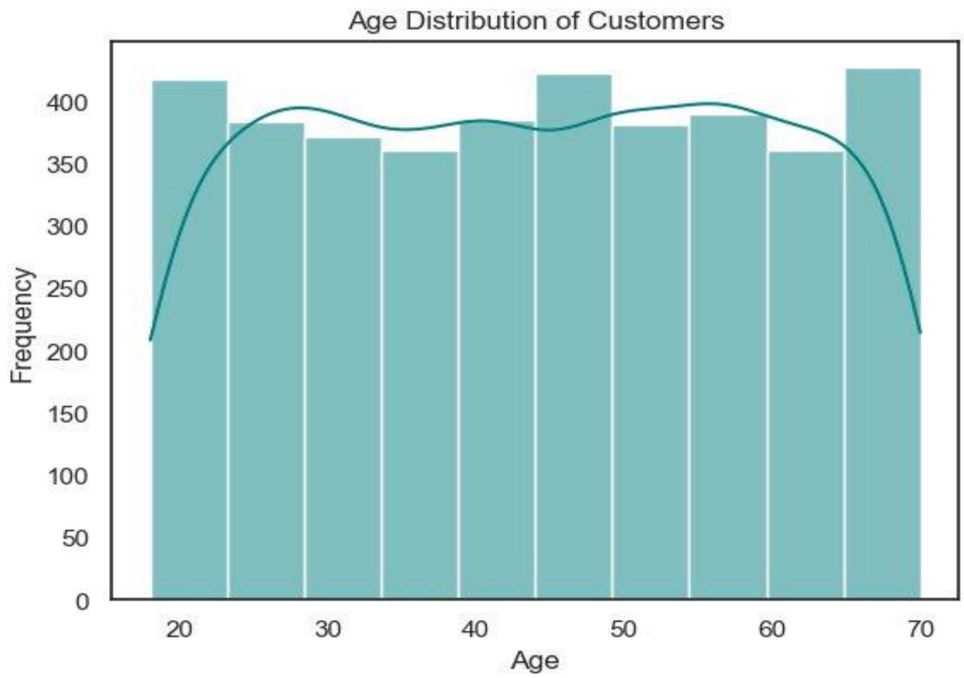
Key Insights

- Every off-diagonal correlation sits between -0.04 and +0.04 — a range so close to zero that it is statistically near-impossible to achieve naturally across five real-world retail variables.
- Purchase Amount has effectively zero correlation with every other variable. Its highest correlation is just 0.031 with Review Rating — confirming nothing in this dataset meaningfully predicts spend.
- Age and Previous Purchases share a correlation of only 0.04. In real retail, younger customers naturally have fewer prior purchases due to shorter tenure — the absence of this signal is a synthetic data fingerprint.
- Review Rating is completely independent of Purchase Amount, Age, and Purchase Frequency — satisfaction scores appear randomly assigned rather than driven by any measurable characteristic. Customer ID shows near-zero correlations with everything — confirming it is a pure identifier with no embedded sequence bias, as expected from deliberate randomisation.

Variable Pair	r	Interpretation
Purchase Amount ↔ Review Rating	0.031	Near zero — satisfaction does not predict spend
Purchase Amount ↔ Age	-0.010	Effectively zero — age irrelevant to spend
Purchase Amount ↔ Prev. Purchases	0.008	Near zero — loyalty does not drive basket size
Age ↔ Previous Purchases	0.040	Near zero — loyalty not age-stratified
Review Rating ↔ Age	-0.022	Near zero — satisfaction independent of age

Visual 10 — Age Distribution of Customers

Histogram with KDE overlay showing the frequency distribution of customer ages. Reveals demographic concentration and identifies whether the dataset reflects a natural population profile.



Key Insights

- The age distribution is remarkably flat — nearly every age bin sits between 350–420 customers with no natural demographic skew. Real retail customer bases never distribute this evenly across a 50-year age span.
- The 18–20 and 65–70 age brackets show the sharpest drop-offs — the only believable edges in the data, likely reflecting genuine population and shopping behaviour limits.
- Two subtle peaks appear at ~20 and ~45–50 in the KDE curve — possibly the only remnants of real demographic demand patterns surviving in an otherwise synthetically balanced distribution.
- The near-uniform distribution severely limits the validity of any age-based segmentation analysis — cross-tabulations with spend, category, or payment method will always return flat results by design.
- The 70+ cutoff is an artificial data boundary, not a behavioural one — the sharp cliff at age 70 is almost certainly a data collection or sampling rule, not a genuine customer behaviour pattern.
- This distribution is the root cause of the flat trends seen throughout the entire analysis — a synthetically balanced age sample suppresses the natural correlations that would emerge in real data.

> **Root Cause Identified**
>
> *The synthetic uniformity of age bands is the primary analytical limitation of this dataset. It suppresses all natural demographic correlations. All prior insights about age, spend, and behaviour must be re-validated against a real, unbalanced customer dataset before any business decisions are made.*

Key Findings & Recommendations

Confirmed Findings

- Clothing is the dominant revenue category across all seasons, sizes, and metrics.
- Outerwear consistently underperforms on revenue, volume, seasonal demand, and stock depth across every chart.
- Spring is the strongest overall trading season — the optimal window for promotional and inventory investment.
- Medium (M) is the top size in every category — stockouts carry the highest risk here.
- Accessories provides the most seasonally stable revenue contribution.
- No tested variable (age, gender, loyalty, discounts, promos, subscriptions, payment method) predicts individual transaction value.

Immediate Actions

Priority	Action	Rationale
High	Audit Outerwear category	Underperforms across all 10 visuals — product, pricing or visibility problem
High	Re-evaluate discount & promo ROI	No basket size uplift detected — measure frequency and acquisition instead
High	Spring season focus	Highest total revenue season — concentrate promotional and inventory budget
Medium	Review size S stock in Clothing	L-to-S imbalance may cause stockouts and missed conversions
Medium	Maintain all payment methods	No method dominant enough to rationalise removing any option
Low	Validate all insights on real data	Synthetic dataset — findings are directional only

Analytical Next Steps

- Build a multivariate model (income, category affinity, geography) to identify real spend drivers beyond age, gender, and loyalty.
- Conduct Customer Lifetime Value (CLV) analysis — determine if subscription and loyalty programmes drive long-term frequency gains even without basket size uplift.
- Run a controlled A/B experiment measuring discount impact on purchase frequency, not just transaction amount.
- Normalise geographic revenue by state population before drawing any location-based investment conclusions.
- Source or generate a real, non-synthetic transactional dataset to validate all directional hypotheses from this analysis.

— End of Report —